Unit-04
CAPSULES

Department of pharmaceutics
CHALAPATHI INSTITUTE OF PHARMACEUTICAL SCIENCES

HARD GELATIN CAPSULE

- DEFINITION:-
  These are solid dosage form of medicaments, in which drug is enclosed within the shells made up of gelatin.

CAPSULE SHELL

These capsules are made up of gelatin blends, small amount of certified dyes, opaquants, plasticizers and preservatives.

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Introduction

- Capsules are solid dosage form in which one or more medicaments are enclosed in a water-soluble, biodegradable shell made up of gelatin.
- Hard gelatin capsule is also referred as “dry filled capsule” (D.F.C.) as their shell is hard and consist of two sections used for filling dry materials in body capsule.

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Advantages of capsules for oral administration

1. Easy to swallow due to their smooth and slippery nature.
2. Easy to handle and carry.
3. Can mask the unpleasant taste, colour and odour of drug using tasteless shell.
4. Better bioavailability than tablets and faster onset of action than tablets.
5. The shells are physiologically inert and easily and quickly digested in the gastrointestinal tract.
6. The shells can be opacified (with titanium dioxide) or coloured, to give protection from light.

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Certified Dyes

Colour approved by D & C act.

Opacifiers

Tio2

Plasticizers

Sorbitol, Glycerin.

Preservatives

Propyl and Methyl Parabens

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GELATIN

- Gelatin is heterogeneous product derived by hydrolytic extraction of animal's collagen.
- The sources of gelatins including animal bones, hide portions and frozen pork skin.

TYPES OF GELATIN

Type A

Type B

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There are two basic types of gelatin

TYPE A

Derived from acid treated precursor that exhibits an iso electric point at pH-9. It is manufactured mainly from pork skin.

TYPE B

Derived from alkali treated precursor that exhibits an iso electric point at pH-4.7. It is manufactured mainly from animal bones

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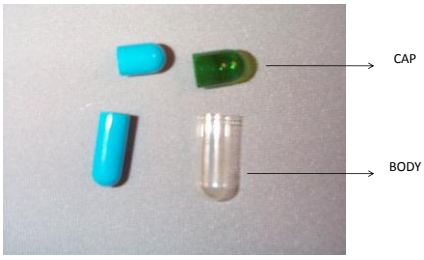
PARTS OF CAPSULE



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MANUFACTURE OF EMPTY GELATIN CAPSULES

Steps involved in making empty gelatin capsules…

- **Dipping**
- **Spinning**
- **Drying**
- **Stripping**
- **Trimming and Joining**
- **Polishing**

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Dipping :

Pairs of the stainless steel pins are dipped into the dipping solution to simultaneously form the caps and bodies.

The dipping solution is maintained at a temperature of about 50^0C in a heated, jacketed dipping pan.

Spinning :

The pins are rotated to distribute the gelatin over the pins uniformly and to avoid the formation of a bead at the capsule ends.

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Drying :

The gelatin is dried by a blast of cool air to form a hard shells.

The pins are moved through a series of air drying kilns to remove water

Stripping :

A series of bronze jaws strip the cap and body portions of the capsules from the pins.

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Trimming and joining

The stripped cap and body portions are trimmed to the required length by stationary knives.

After trimming to the right length, the cap and body portion are joined and ejected from the machine.

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SIZE OF CAPSULES

Size	Volume in ml	Size in mm
000	1.37	26.3
00	0.95	23.7
0	0.68	21.8
1	0.50	19.2
2	0.37	18.3
3	0.30	15.3
4	0.21	14.7
5	0.15	11.9

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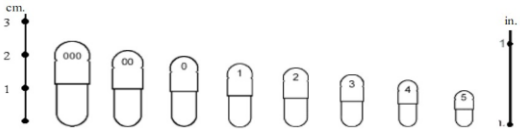
SHAPES OF CAPLUSE



*The largest size of the capsule is No: 000.
*The smallest size is No: 5.
*The standard shape of capsules is traditional, symmetrical
bullet shape.

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Filling hard capsule shells

- Hand-operated capsule filling machines or Feton capsule filling machine
- Consist of a couple of plates are capable of producing about 200 to 2000capsules/hr.

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A hand operated gelatin capsule filling machine consists of the following parts : -

1. A bed with 200-300 holes.
2. A capsule loading tray
3. A powder filling tray
4. A pin plate having 200 or 300 pins corresponding to the number of holes in the bed and cap holding tray.
5. A lever handle
6. A Cam handle
7. A plate fitted with rubber top.

Filling hard capsule shells

The process of working:

- Rectification
- Separating the caps from empty capsules
- Filling the bodies
- Scraping the excess powder
- Replacing the caps
- Sealing the capsules
- Cleaning the outside of the filled capsules
- 160,000 capsules per 8hour shift

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Rectification

- The empty capsules are oriented so that all point the same direction i.e. body end downwards
- In general, capsule pass one at a time through a channel just wide enough to provide grip at cap end
- The capsules will always be aligned body end downwards, regardless of which end entered the channel first with the help of specially designated blades

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Separation of caps from body

- The rectified capsules are delivered body end first into the upper portion of split bushings or split filling rings
- A vacuum applied from below pulls the body down into the lower portion of the split bushing
- The diameter of the bush is too large to allow them to follow body
- The split bushings are separated to expose the bodies for filling

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FILLING OF HARD GELATIN CAPSULES

Various Filling Machine Available…

- Eli-lily and Co
- Farmatic
- Hofliger and Karg
- Zanasi
- Parke-Davis.

- These machine differ in their design and output

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Cleaning and polishing capsules

Small amount of powder may adhere to the outside of capsules after filling.

1) Salt Polishing- NaCl
2) Cloth Dusting.
3) Brushing.
4) Pan Polishing- Acela-cota pan is used to dust and polish.

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CAPSULE POLISHING MACHINE

AUTO MATIC CAPSULE ARRANGEMNT

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Filling

- **Auger fill principle**: Because the auger mounted in the hopper rotates at a constant rate, the rate of delivery of the powder to the capsules tend to be constant
  - Flat blade auger
  - Screw auger

    E.g. Capsugel type 8 filling machine

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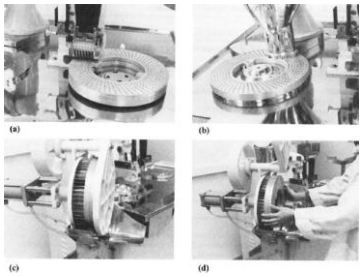
Semiautomatic capsule-filling machine. (a)``Sandwich'' of cap and body rings positioned under rectifier to receive empty capsules. Vacuum is pulled from beneath the rings to separate caps from bodies. (b) Body ring is positioned under foot of powder hopper for filling. (c) After filling the bodies, the cap and body rings are rejoined and positioned in front of pins. These pins push the bodies to engage the caps. (d) The plate is swung aside and the pins are used to eject the closed capsules



Filling

- **Vibratory fill principle**: In the powder, a perforated resin plate is positioned and connected to a vibrator
- The powder blend tends to be fluidized by the vibration of plate and assists the powder to flow into the bodies through the holes in resin plate

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OSAKA MODEL R-180
SEMI AUTOMATIC CAPSULE
FILLING MACHINE

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Filling

- **Piston tamp principle**:
- In this pistons or tamping pins lightly compress the individual doses of the powders into plugs(also called as slugs) and eject the plugs into empty capsule bodies
  - DOSATOR PRINCIPLE
  - DOSING DISC PRINCIPLE

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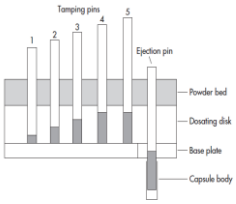
Filling

- **Piston tamp principle**:
  - DOSING DISC PRINCIPLE: a solid brass 'stop' plate is sliding down the dosing disc to close off the hole. Five sets of pistons compress the powder into cavities to form plugs



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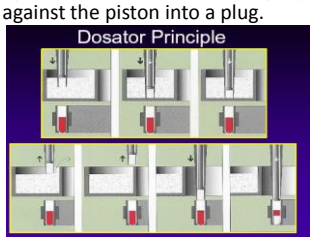
- DOSATOR PRINCIPLE: it consists of cylindrical dosing tube fitted with movable piston. The position of the piston is preset to a particular height to define a volume. Powder enters the open end of dosator and is slightly compressed against the piston into a plug.



HOFLIGER KARG AUTOMATIC CAPSULE FILLING MACHINE

ZANASI AUTOMATIC CAPSULE FILLING MACHINE

Filling of hard gelatin capsules

✓ Equipment used are: -

- **Hofliger-Karg machine:**
  - ➢ Formation of compacts in a die plate using tam ping pins to form a compact.
- **Zanasi or Martelli encapsulator:**
  - ➢ Forms slugs in a dosator which is a hollow tube with a plunger to eject capsule plug.

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Storage, packaging, and stability

- Finished capsules normally contain an EMC of 13-16%.
- < 12% MC, the capsule shells become brittle
- >18% make them too soft
- To maintain a relative humidity of 40-60% when handling and storing capsules.
- QUALI-V, developed by Shionogi Qualicaps, is the first HPMC capsule developed for eventual use in pharmaceutical products.

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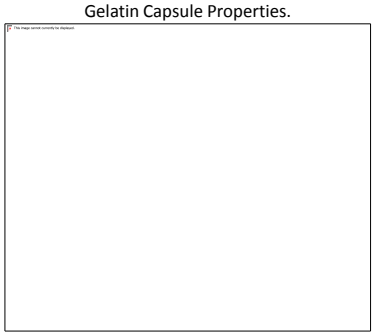
Relative Humidity (RH), Gelatin Moisture Content, and Hard Gelatin Capsule Properties.



FILLING OF HARD GELATIN CAPSULES

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ZANASI AUTOMATIC CAPSULE FILLING MACHINE

HOFLIGER KARG AUTOMATIC CAPSULE FILLING MACHINE

Polishing

Pan Polishing : Acela-cota pan is used to dust and polish.

Cloth Dusting : Capsule are rubbed with cloth.

Brushing : Capsule are feed under soft rotating brush.

Storage

Finished capsules normally contain an equilibrium moisture content of 13-16%.
To maintain a relative humidity of 40-60% when handling and storing capsules

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SOFT GELATIN CAPSULE

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SOFT GELATIN CAPSULE

Definition:-

Soft Gelatin capsules are one piece, hermetically sealed, soft gelatin shells containing a liquid, a suspension, or a semisolid.

Soft gelatin is mainly composed of gelatin, plasticizers, preservative, colouring and opacifying agents, flavoring agents and sugars.

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APPLICATION OF SOFT GELATIN CAPSULE:

The pharmaceutical applications of soft gelatin capsules are:

- as an oral dosage form
- as a suppository dosage form
- as a specialty package in tube form, for human and veterinary use, single dose application for topical, ophthalmic, and rectal ointments.

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Plasticizer and Gelatin ratio

In soft gelatin capsule the amount of plasticizers used is more

In soft gelatin capsule the plasticizer and gelatin ratio is

0.8 : 1

In hard gelatin capsule the plasticizer and gelatin ratio is

0.4 : 1

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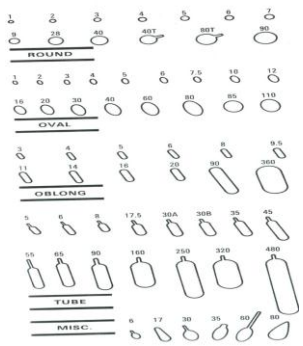
SHAPE OF CAPSULE

The shape of soft gelatin capsule are round, oval, oblong, tube.



MANUFACTURE OF SOFT GELATIN CAPSULES

Is manufactured by four methods

- Plate process
- Rotary die process
- Reciprocating die
- Accogel machine

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Plate process:

- Place the gelatin sheet over a die plate containing numerous die pockets,
- Application of vacuum to draw the sheet in to the die pockets,
- Fill the pockets with liquid or paste,
- Place another gelatin sheet over the filled pockets, and
- Sandwich under a die press where the capsules are formed and cut out.

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Rotary die process:

The material to be encapsulated flows by gravity. the gelatin sheets are feed on rolls contain small orifice lined up with the die pocket of the die roll.

Two plasticized gelatin ribbons are continuously and simultaneously fed with the liquid or paste fill between the rollers of the rotary die mechanism where the capsule are simultaneously filled, shaped, hermetically sealed and cut from the gelatin ribbon.

The sealing of the capsule is achieved by mechanical pressure on the die rolls and the heating(37-40°C) of the ribbons by the wedge.

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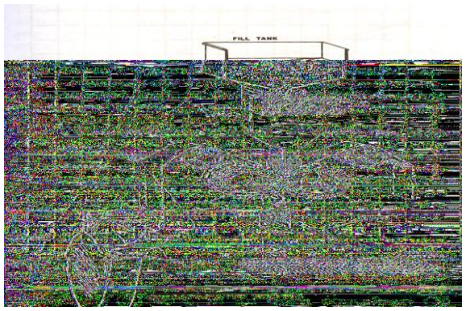
The rotary die process



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IMPORTANT SPECIFICATIONS OF GELATIN

Bloom or gel strength: It is a measure of cohesive strength of cross-linkage that occurs between molecules and is proportion to the molecular weight of gelatin.

Bloom is determined by measuring the weight in grams required to move a plastic plunger of 0.5inches in diameter, 4mm into a 62/3% gelatin that has held at 10°C for 17 hrs.

The unit of bloom is grams and it is between 150-250g

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Viscosity: Is determined on a 62/3% gelatin of water at 60°C and it is a measure of the molecular chain length.

Standard used: 25-45 mill poise.

Iron content: Iron is always present in raw gelatin, and its concentration usually depends on the iron content of the large quantities of water used in its manufacture .

amount should not exceed 15ppm.

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QUALITY CONTROL OF CAPSULES

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QUALITY CONTROL OF CAPSULES

Whether capsules are produced on a small scale or large scale all of them are required to pass through certain tests i.e., quality control tests to test the quality of the finished product.

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Quality control tests are divided into

PHYSICAL TEST

- Disintegration test
- Weight variation

CHEMICAL TEST

- Dissolution test
- Assay
- Content uniformity
- Stability testing
- Moisture permeation test

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PHYSICAL TESTS OF CAPSULES

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Finally physical control processing and packing may be accomplished by the following in line continuous operations

1.A capsule diameter sorter allows to pass to the next unit of any capsule with in + or _ 0.020 inch of theoretical diameter .

2.A capsule colour -

the capsules are fed to it automatic from the diameter sorter by a pneumatic conveyer .In this unit, any capsule whose colour does not conform to the reference colour standard for that particular product is discarded others passes the test.

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DISINTERATION TEST-

- The disintegration test determines the whether capsules disintegrated with a prescribed time when placed in a liquid medium under the prescribed integral conditions .

METHOD-

- According to B.P and which applies to both hard and soft capsules

1.introduce one capsule in each tube and suspend the apparatus in a beaker containing 60ml water at 37^0 C,

– if hard capsules float on surface of water, the disc may be added.

2.Operate the apparatus for 30 min, remove the assembly from the liquid.

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3.the capsule pass the test if

- No residue remains on the screen of the apparatus or,
- If the residue remains, it consists of fragments shells ,
- If a soft mass with no palpable core ,
- If the disc is used any residue is remaining on its lower surface should only consists of fragments of shells.

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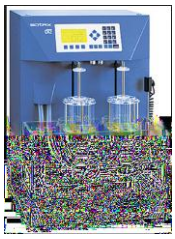

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WEIGHT VARIATION

FOR HARD CAPSULES

weigh 20 capsules individually and determine the avg weight

↓

The individual wts should be with in limit of 90-110% of avg wt

↓

If not all of capsules fall with in the limits,

Weigh 20 capsules individually

↓

Remove the net content of each capsule with the aid of a small brush

↓

Weigh the empty shells individually

↓

NET WT OF CONTENTS INDIVIDUALLY = THE WT OF SHELL-GROSS WT

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Determine the avg net content from the sum of individual net wt

↓

Then determine the difference b/w each individual net content and avg net content

LIMITS-

- Not more then 2 of the differences are greater then 10% of the avg net content
- No case is the difference greater then 25% wt range

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If more then 2 ,but not more then 6 capsules deviate from the avg b/w 10-25%

↓

Determine the net contents of an additional 40 capsules

↓

Determine the avg content of entire 60 capsules

↓

Determine the 60 deviations from the new avg

LIMITS-

- NMT 6 of 60 capsules does the difference exceed 10% of the avg net content
- No case does the difference exceed 25%

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FOR SOFT CAPSULES

proceed as directed under hard capsules, but determine the net wt of the contents of individual capsules as follows:

weigh the capsules individually then cut and open the capsules

↓

remove the contents by washing with the suitable solvent

↓

allow the solvents to evaporate from the shells at room temp

↓

weigh the individual shells

↓

Calculate the net contents

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CHEMICAL TESTS OF CAPSULES

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DISSOLUTION TEST-

- The dissolution test is carried out using the dissolution apparatus official in both the U.S.P and I.P .
- The capsule is placed in a basket , and the basket is immersed in the dissolution medium and caused to rotate at a specified speed .
- The dissolution medium is held in a covered 1000ml glass vessel and maintained at 37^0 c + 0.5 ^{0c} by means of a constant temperature suitable water bath .
- The stirrer speed and type of dissolution medium are specified in the individual monograph .

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RESULT -

- Six capsules are tested and are accepted if each of them is not less than monograph specified i.e., p +5%
- If it fails then additional six capsules are tested the result is accepted if the avg. of 12 capsules is greater than or equal to p and none of them is less than p-15%.
- If the capsule still fails the test the additional 12 capsules are tested and are accepted if the avg. of 24 is greater than to p, if not more than two less than p-15% and none of them is less than p-25%

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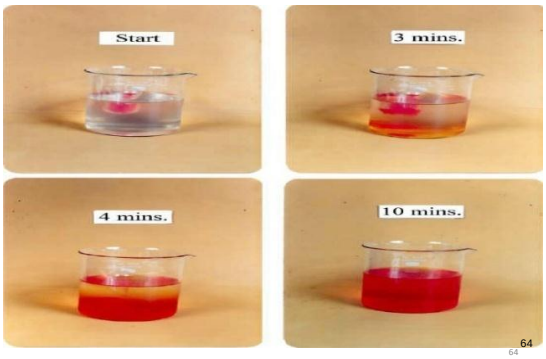
DISSOLUTION PROFILE



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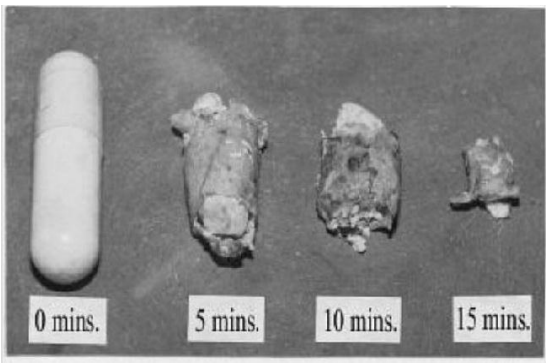

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FACTORS AFFECTING DRUG DISSOLUTION FROM HARD GELATIN CAPSULES

Overall Dissolution Rate is a Function of:

- Dissolution Rate of the Shell
- Rate of Penetration of Dissolution Medium
- Rate of Deaggregationof Powder Mass
- Nature of Primary Drug Particles

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- Normally, shell ruptures and dissolves within about 4 minutes.
- Rupture occurs first at the shoulders where shell wall is thinnest.
- Ends fall away and as liquid penetrates and deaggregation occurs, formulation tend to spill out of the two ends.

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CONTENT UNIFORMITY

10 capsules are taken and subjected to assay

↓

9 of 10 capsules should be in the range of +-15% (85-115%)

↓

And 10th capsule are beyond +_ 15% range then 20 capsules are assayed

↓

All capsules with in range of +_25% (75-125%)

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MOISTURE PERMEATION TEST

The degree and rate of moisture penetration is determined by packaging the dosage unit together with a colour revealing desiccant pellet

↓

Expose the packed unit to known relative humidity over a specified time

↓

Observe the desiccant pellet for colour change

↓

Any change in colour indicates absorption of moisture

↓

By measuring pre test weight and protest weight of pellet, amount can be calculated.

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BLOOM STRENGHT OF GELATIN

RAW MATERIALS-

The gelatin of the capsule shells should be assayed for varies physical properties like bloom strength ,viscosity etc..

PROCEDURE-

gelatin is weighed into water to typically create a 6.67% sol^{n} in standard bloom bottles

↓

The mix is then stirred and keep it for 3 hours at room temp

↓

Bottles are placed in a 65^{0} c bath for 20 minutes

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Allow the bloom jars to cool for 15 min at room temp

↓

They are then conditioned for 16 hrs in 10^{0} c water bath

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when conducting gelatin bloom test,

the bloom jar is centred with the probe just above the sample surface

↓

The probe penetrates the gelatin to a target depth of 4mm at a speed of 0.5mm/s , and then retracts

↓

The peak force is the gel strength in grams bloom

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- The physical stability of soft gelatin capsules is associated primarily with the pick up or loss of water by the capsule shell .
- If these are prevented by proper packaging ,the above controlled capsule should have satisfactory physical stability at S temp ranging from just above freezing to as high as 60^0 c.
- As the humidity increases the moisture content pickup of capsules increases .

  ex- at 30%RH at room temp shows that gelatin retain about12%(48 mg) of water and glycerin 7%(14 mg)of water.

  at 60%RH the moisture content should be 17.4%.
- High humidity (>60%RH at 21-24^0 c)produce more lasting effects on the capsule shell.

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Since as moisture is absorbed ,the capsules become softer ,tackier , and bloated.

- The capsule manufacturer routinely conducts accelerated stability tests on new product as an integral part of the production development program.
- The successful results are obtained by conducing at test conditions like

  1.80%RH at room temp in an open container

  2.40^0 c in open container

  3. 40^0 c in closed container (glass bottle with tight screw cap)

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chemists conducting the physical stability test in their own lab should keep two important points in mind

1.prior to testing ,the capsule should be equilibrated to known atm conditions, preferably 20-30%RH at 21-24^0 c.

2.evaluation of the results of the previously described heat test should be made only after the capsules have returned to equilibrium to room temp

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Capsule stability

- Stability:20 to 30 % RH at 21 to 24 ° C

Table 2: Effect of Temperature and Humidity on Capsule shell

Temperature	Humidity	Effect on Capsule shell
21-24°C	60%	Capsules become softer,tackier and bloated
Greater than 24°C	Greater than 45%	More rapid and pronounced effects – unprotected capsules melt and fuse together

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PACKAGING OF CAPSULES

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PACKAGING AND STORAGE OF CAPSULES

Capsules should be packed in a well-closed glass or plastic containers and stored in a cool place.

- These type of containers have advantage over cardboard boxes that they are more convenient to handle and transport and protect the capsules from moisture and dust.
- To prevent the capsules from rattling a tuft of cotton is placed over and under the capsules in the vials.
- In vials containing very hygroscopic capsules a packet-containing desiccant like silica gel or anhydrous calcium chloride

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may be placed to prevent the absorption of excessive moisture by the capsules.

- Now a days capsules are strip packaged which provide sanitary handling of medicines, ease in counting and identification .

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- Plastic bottle with screw cap
(most popular package in USA)

- Clam shell blister (one piece plastic that folds over and locks itself; no heating required)

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- Blister pack (heat sealed blister on a cardboard)

- Plastic pail/bucket(economical bulk package)

- Plastic pouch zip locked (for sale via retail stores or route trucks must be packed in outer case for shipping)

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